

Narrative Chain-of-Thought: Inference-Time Scaffolding for Defeasible Ethical Reasoning in Large Language Models

Anonymous ACL submission

Abstract

Standard chain-of-thought on moral dilemmas exhibits two structural failure modes that matter for deployment: stakeholder collapse, where the model reduces a multi-party situation to a binary, and uncertainty suppression, where the model commits to a projected outcome it cannot warrant. We introduce narrative chain-of-thought (N-CoT), a system prompt that asks the model to name a protagonist, enumerate stakeholders, project two-step consequences, articulate uncertainty, and only then commit. The protocol adds no training data, parameters, or fine-tuning. On a 100-scenario DailyDilemmas sample across four generators from three vendors, N-CoT cuts stakeholder collapse from up to 31% to under 1% and uncertainty suppression from up to 72% to 1–24% on every model. A section-by-section ablation attributes each structural shift to the specific sub-instruction that carries it, with negligible cross-variable spillover. Extended to a four-round multi-stakeholder protocol with a moderator-built integrated proposal and a binary accept/reject vote, the same scaffold converts a 6% debate standoff into 95% full consensus while preserving a small fraction of principled structural rejections in the roles whose stake the integrated proposal materially undermines. The intervention is a small change to the prompt that removes two failure modes firing near-universally on current frontier models. The resulting traces externalise the stakeholders, projected consequences, and uncertainty that ground each commitment, and the multi-agent extension preserves principled dissent under integrated proposals, providing an auditable substrate for dependable agentic-workflow deployment.

1 Introduction

Consider a familiar family question. An aging parent has had two minor accidents in the past year and refuses to give up driving. Should the family

insist? A frontier LLM prompted to think step by step on this scenario typically picks a side within a few sentences, projects a confident downstream outcome (“safety first”; “they will adjust”), and rarely names the people whose lives the decision actually touches: the parent’s autonomy and social access, the pedestrians on familiar routes, the relatives who will inherit the conflict, the friends who depend on the parent for rides. The model is not necessarily wrong; the trace it produced does not look like the kind of reasoning the decision warrants.

The pattern is not specific to one dilemma. On the DailyDilemmas corpus (Chiu et al., 2025), four frontier generators spanning three vendors (OpenAI, Anthropic, xAI) and two model tiers suppress articulated uncertainty on 50–72% of standard-CoT outputs and collapse the stakeholder set on 15–31%. This is a structural deficit of the trace itself: the model commits to a projected outcome it cannot warrant, and it does so against a flattened picture of who is affected. The same shape of deficit is coherent with deployment-relevant failures reported elsewhere, including the GPT-4o sycophancy rollback (OpenAI, 2025b,a), agentic-misalignment blackmail rates up to 96% in simulated deployments (Lynch et al., 2025), and reasoning-induced misalignment (Yan et al., 2025; Sharma et al., 2024).

We introduce narrative chain-of-thought (N-CoT), a system prompt that puts the model on the manifold of human narrative its pretraining corpus most heavily samples. The model is asked, in first person, to name a protagonist, enumerate stakeholders, project two-step consequences, articulate uncertainty, and only then commit. Figure 1 states the mechanism.

What we borrow from the multi-agent systems for scientific discovery of Gottweis et al. (2025) and Lu et al. (2024) is one methodological move: scaffold LLM reasoning into the primitives of

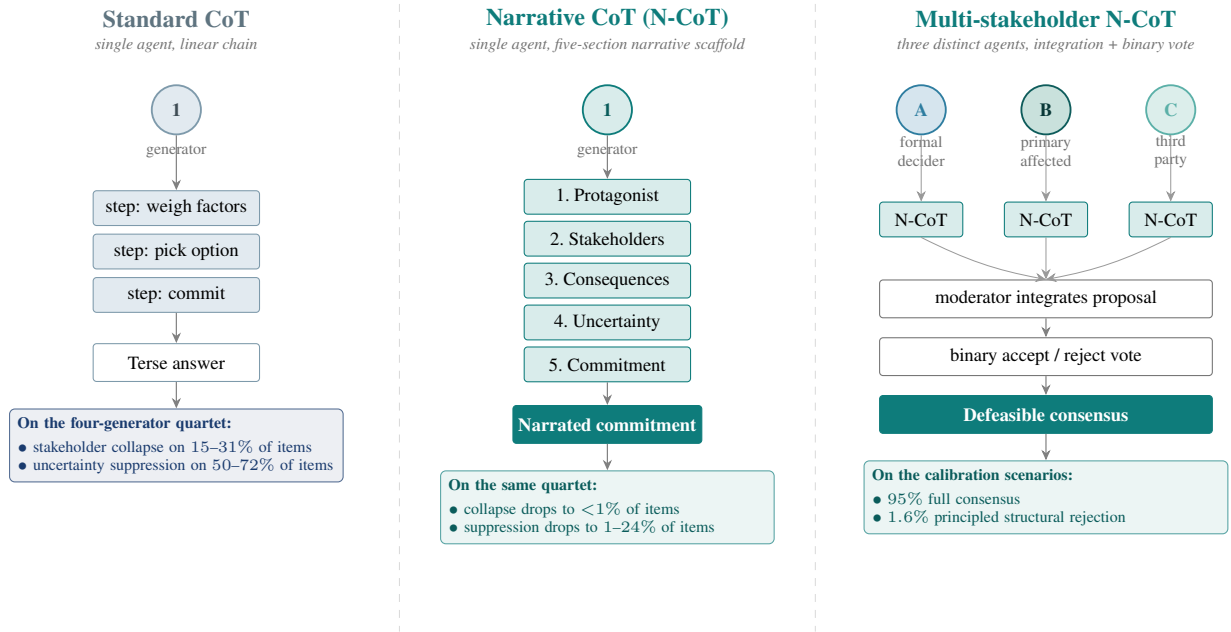


Figure 1: One scaffold, three settings. Standard CoT (left) lets one generator run a linear reasoning chain over the dilemma; the two failure modes diagnosed in §4 fire on this scaffold. Narrative CoT (middle) keeps the single generator but forces the trace through five narrative primitives (protagonist, stakeholders, consequences, uncertainty, commitment) before any commit; on the four-generator quartet this drops stakeholder collapse to <1% and uncertainty suppression to 1–24% of items. Multi-stakeholder N-CoT (right) reuses N-CoT as the per-agent generator and adds three distinct stakeholder narrators (formal decider, primary affected party, third party) whose modification requests are integrated by a moderator and then put to a binary accept/reject vote; on the calibration scenarios this protocol yields 95% full consensus with 1.6% principled structural rejection (§5). The same scaffold composes from one narrator to many.

the target domain, then integrate across primitives with a meta-review step at the top of the hierarchy. Three things distinguish our application of that move. The primitives we reify are narrative rather than scientific (protagonist, stakeholders, consequences, uncertainty, commitment), so the artefact a reader inspects is a coherent story rather than a ranked candidate list. The single-agent layer is a contribution in its own right: §4 shows that reifying these primitives inside one generator already eliminates two cross-vendor failure modes of standard CoT, before any multi-agent composition is invoked. And because ethics has no experimental-falsification oracle, the success criterion that replaces hypothesis ranking is consensus that preserves principled dissent, evaluated in §5 via a binary integrated-proposal vote whose residual rejections concentrate in the roles whose stake the proposal materially undermines.

This paper reports three results. First, on a 100-scenario DailyDilemmas sample across the four-generator quartet, N-CoT cuts both failure modes to floor on every model (§4). Second, a sub-instruction ablation on claude-sonnet-4-6

attributes each structural shift to the specific N-CoT sub-instruction that carries it, with negligible cross-variable spillover (§4.1). Third, extended to a four-round multi-stakeholder protocol with an integrated-proposal vote, the scaffold converts a 6% debate standoff into 95% full consensus while preserving a small 1.6% structural-rejection rate concentrated in roles whose stake the integrated proposal undermines (§5). The intervention is a small auditable prompt change that adds no weights or training data. The resulting trace externalises the stakeholders, the projected consequences, and the uncertainty that ground each commitment, and the multi-agent extension preserves principled dissent rather than collapsing into uniform agreement. We read the combination as evidence that a dependable, inspectable substrate for ethical reasoning is recoverable from current pretrained models by inference-time scaffolding alone, a property well-suited to agentic-workflow deployments where auditable rationale matters as much as the chosen action. The saturated downstream deployment probes that motivated the diagnosis remain saturated under either

scaffold (Appendix E); harder probe sets are pre-registered as future work.

2 Background and Related Work

Chain-of-thought and its failure modes.

Chain-of-thought prompting (Wei et al., 2022; Kojima et al., 2022) elicits step-by-step intermediate reasoning before a final answer; it improves arithmetic and symbolic tasks but does not by construction force the trace to track situational structure. Recent analysis identifies a sharp tension: as chain-of-thought reasoning capacity grows, models become more responsive to harmful requests because reasoning and safety entangle in shared neuron populations, a phenomenon Yan et al. (2025) describe as “reasoning-induced misalignment”. N-CoT can be read as a constrained chain-of-thought that re-anchors the trace to a narrative-causal structure before the reasoning capacity is exercised; §4 shows that the intervention works on a current reasoning model as well as a non-reasoning one.

Narrative scaffolding for reasoning.

Visualisation-of-Thought (Wu et al., 2024) shows that asking a model to externalise reasoning in the native representational format of a domain (spatial diagrams for spatial tasks) improves reliability. We apply the same principle to ethical reasoning: the native format is a story with named agents, projected consequences, and uncertainty (MacIntyre, 1981; Bruner, 1986). Empirical narrative-perspective interventions in medical decision contexts measurably increase perspective-taking and empathic concern (Shaffer et al., 2019; Bientzle et al., 2021, 2024), and recent work on generative agents shows long-horizon coherent behaviour is achievable from narrated character state (Park et al., 2023).

Causal inference from narrative. The Computational Theory of Mind (Fodor, 1975; Putnam, 1967) treats understanding a sequence of events as inferring a structural causal model that explains them; the Kuleshov experiment (Kuleshov, 1974; Bordwell, 1985) is a textbook example. LLMs perform near random baseline on causal inference stripped of empirical pattern matches (Jin et al., 2024): pretraining supplies the ability to pattern-match narrative token sequences, not the structured causal reasoning that determines how those narratives resolve. The role of N-CoT is to convert

that pattern matching into a usable causal trace at inference time.

Multi-agent debate and defeasibility. Irving et al. (2018) proposed AI safety via debate; Du et al. (2023) showed multi-agent debate improves factual reasoning. Both target task-correctness, not stakeholder-divergent value standoffs. Defeasible reasoning, the property that conclusions can be revised in light of new information without abandoning the underlying inference scheme, has a long lineage in moral philosophy (Pollock, 1987; MacIntyre, 1981). The protocol we evaluate in §5 stacks defeasibility on top of N-CoT and asks whether present-day models can be elicited to revise narrated commitments under narrated counter-evidence.

Multi-agent LLM systems for scientific discovery. A growing line of work scaffolds LLM reasoning into the primitives of a target domain. The AI co-scientist of Gottweis et al. (2025) reifies hypothesis generation, peer review, ranking, evolution, and meta-review as co-operating agents; the AI Scientist (Lu et al., 2024) closes the loop end-to-end; Coscientist (Boiko et al., 2023), ChemCrow (M. Bran et al., 2024), Agent Laboratory (Schmidgall et al., 2025), and the Virtual Lab (Swanson et al., 2025) instantiate the move in chemistry and biomedicine. Our protocols ask the analogous question one level down: can a scaffold that reifies the primitives of human ethical reflection deliver the same kind of inference-time quality lift on moral reasoning that scientific-method scaffolds deliver on hypothesis generation? The moderator’s integrated-proposal step in §5 is structurally the meta-review of Gottweis et al. (2025), applied to value-laden positions. We make this substitution because the absence of a ground-truth oracle in ethics requires a different multi-agent operation: stakeholder positions cannot be ranked the way competing scientific hypotheses can be ranked against eventual experimental falsification, so the right move is integration-then-veto rather than ranking-then-selection, turning the residual veto pattern itself into the measurement of interest.

3 Narrative Chain-of-Thought

Narrative chain-of-thought (N-CoT) is a system prompt that instructs the model to produce, in first person, a five-section trace before any final an-

Algorithm 1 Narrative chain-of-thought (N-CoT): generation and structural coding for a single dilemma.

Require: dilemma s ; generator G ; judge J ; decoder Θ
Ensure: trace t and structural codes c

- 1: **Build N-CoT system prompt** π_{NCOT} requesting five first-person sections:
- 2: (1) PROTAGONIST: name role and epistemic state
- 3: (2) STAKEHOLDERS: parties intersecting the decision
- 4: (3) CONSEQUENCES: each action ≥ 2 steps forward
- 5: (4) UNCERTAINTY: what remains genuinely unknown
- 6: (5) COMMITMENT: chosen action and narrative warrant
- 7: **Generate trace**
- 8: $t \leftarrow G(\pi_{\text{NCOT}} \parallel s; \Theta)$
- 9: **Code structural variables** with judge J ’s rubric
- 10: $(sc, mh, us, nf, ref, tr) \leftarrow J(t)$
- 11: *// stakeholder count, max causal hops, uncertainty score,*
- 12: *// framework count, refused flag, truncated flag*
- 13: **Derive failure-mode tags** (deterministic)
- 14: $\text{STAKEHOLDERCOLLAPSE} \leftarrow sc \leq 1$
- 15: $\text{UNCERTAINTYSUPPRESSION} \leftarrow us = 0$
- 16: $\text{CONSEQUENTIALFLATTENING} \leftarrow mh \leq 1$
- 17: $\text{FRAMEWORKENUMERATION} \leftarrow nf \geq 2$
- 18: $\text{PREMATUREREFUSAL} \leftarrow ref$
- 19: **return** (t, c)

swer: (1) characterise the protagonist (name, role, what they know); (2) enumerate the stakeholders whose lives intersect the decision and what is at stake for each; (3) project the consequences of each available action at least two steps out for each stakeholder; (4) state what remains uncertain about each projected future; (5) commit to a decision and explain why, within the narrative frame, that trajectory is preferable to the alternatives.

The intervention is a single change to the system prompt; it adds no training data, parameters, or fine-tuning, and it does not mention causal graphs, utilities, or any formal apparatus. The mechanism it exploits is that text matching the five-section structure is densely represented in the narrative subdistribution of the pretraining corpus, so the model can amortise causal-trajectory inference by retrieval rather than by reasoning from first principles. Algorithm 1 states the procedure.

3.1 Multi-Stakeholder Narrative Deliberation

A single N-CoT trace commits one protagonist to a position; many real decisions involve multiple narrators with conflicting stakes. We extend N-CoT into a four-round protocol that re-uses N-CoT as the per-agent generator. For each scenario, three stakeholder perspectives (formal decider, primary affected party, third party) each produce

an N-CoT statement (Round 0), exchange rebuttals (Round 1), restate a final position (Round 2), respond to a moderator-built synthesis with ACCEPT, ACCEPT_WITH_MODIFICATION, or REJECT (Round 3), and cast a binary ACCEPT/REJECT vote on a single integrated proposal the moderator constructs from the three modification requests (Round 4). The integration step in Round 4 makes defeasibility measurable: an agent who continues to reject a proposal that addresses its modifications is signalling a structurally non-negotiable concern, not procedural friction.

4 Experiment 1: N-CoT at Scale

Setup. Three prompting conditions are compared: bare input/output, standard CoT (“Think step by step, then give your answer”), and N-CoT (the five-section structure of §3, with no formal apparatus mentioned). Decoding parameters are held identical across conditions. Four generators are tested: gpt-5.4-nano ($N=20$) from the original pilot, plus three new generators spanning two additional vendors: claude-haiku-4-5 ($N=20$), grok-4-1-fast-reasoning ($N=5$), and the flagship claude-sonnet-4-6 ($N=5$, cost-controlled). Here N is the number of independent samples per (scenario, condition, generator) cell drawn at temperature 1; with the 100-scenario sample and three conditions this yields $300N$ generations per generator before the cross-judge coding stage. Coded sample sizes (N_S, N_N in Table 1) are slightly smaller because the rubric drops generations where the judge could not extract a coded variable. Outputs are coded by two cross-vendor judges (claude-haiku-4-5 primary, gpt-5.4-nano secondary) on a six-variable rubric with quadratic-weighted Cohen’s κ (Cohen, 1960) per variable. The scenario pool is a 100-scenario stratified sample of DailyDilemmas (Chiu et al., 2025), spanning 18 topic groups; the same deterministic stratified sample (seed 42) is used for all four generators, making the scenario sets comparable.

N-CoT eliminates both failure modes on every model. Uncertainty suppression fires on 50.0–72.3% of standard-CoT outputs across the quartet (Table 1); stakeholder collapse on 14.6–30.6%. Both collapse under N-CoT: stakeholder collapse to 0.0–1.2%, uncertainty suppression to 0.8–24.5% (Figure 2). The reduction is largest where the standard-CoT rate is largest, and the pattern is

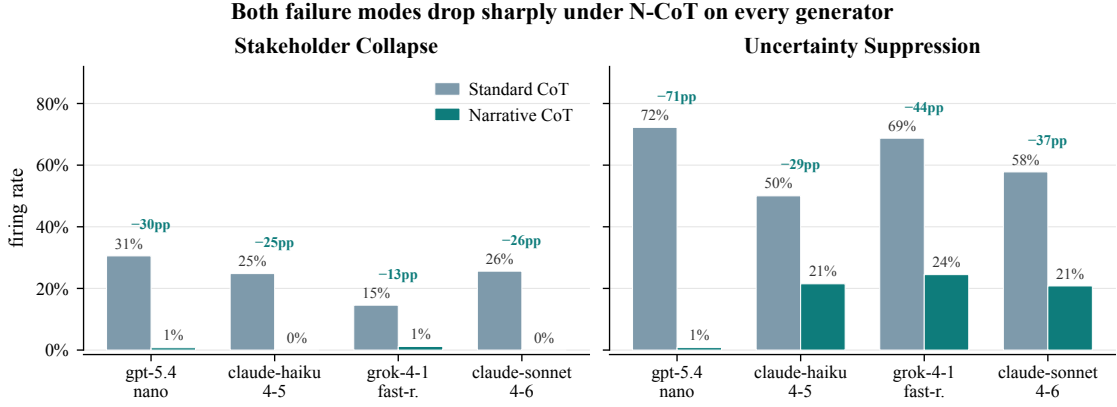


Figure 2: Failure-mode firing rates (standard CoT vs. N-CoT) across the four-generator quartet (gpt-5.4-nano, claude-haiku-4-5, grok-4-1-fast-reasoning, claude-sonnet-4-6). The suppression pattern from the original pilot replicates across vendors and model tiers; uncertainty suppression drops by 28–72 percentage points and stakeholder collapse drops to near zero on every model.

Generator (vendor, tier)	N_s	N_N	Δ_{sc}	Δ_{us}
gpt-5.4-nano (OpenAI, b)	1,989	1,985	−30	−72
claude-haiku-4-5 (Anth., b)	1,999	1,996	−25	−29
grok-4-1-fast-r. (xAI, b)	515	515	−13	−44
claude-sonnet-4-6 (Anth., f)	500	500	−26	−37

Table 1: Cell sizes and percentage-point drops for the two failure modes that empirically fire (Δ_{sc} = stakeholder collapse, Δ_{us} = uncertainty suppression; both N-CoT minus standard CoT, in percentage points; N_s , N_N are coded samples under standard CoT and N-CoT respectively). Vendor tiers are b = budget, f = flagship. Figure 2 plots the raw rates with the same drop annotations. Premature refusal, framework enumeration, and consequential flattening fire below 1% on all generators and are reported as untested.

log(output length) pooled across conditions and re-evaluating δ on residuals, the structural shift survives length removal on the OpenAI and xAI generators (gpt-5.4-nano $\delta_{resid} = +0.51$ on stakeholder count and $+0.77$ on uncertainty score; grok-4-1-fast-reasoning $+0.33$ and $+0.43$; all CIs strictly above zero). On the two Anthropic generators the residualised effect is statistically indistinguishable from zero, which we read as those models already exhibiting the target structural density per unit length under standard CoT, so the prompt-side gain there operates through additional length rather than additional per-unit density. Per-generator δ s with bootstrap CIs are in Appendix B (Table 5, Figure 5).

4.1 Sub-Instruction Ablation

To identify which section of the N-CoT prompt carries the structural shift, we run a sub-instruction ablation on claude-sonnet-4-6 ($N=3$, 30-scenario stratified subsample, seed 43; cost-controlled). Six conditions are compared: the full N-CoT prompt (control) and five variants each dropping exactly one of the five sections. Figure 3 reports Cliff’s deltas for each drop condition relative to the full N-CoT control on the two headline structural variables.

The ablation shows direct causal attribution. Dropping Stakeholders reduces stakeholder count by $\delta = -0.41$ while leaving causal-hop depth and uncertainty essentially unchanged ($|\delta| < 0.03$); dropping Uncertainty reduces uncertainty score by $\delta = -0.92$, nearly the maximum possible, while leaving every other variable untouched; dropping Consequences reduces causal-hop depth by $\delta =$

cross-vendor: no single vendor escapes the baseline, and no single vendor escapes the intervention. gpt-5.4-nano collapses both modes to $<1\%$. The two Anthropic generators retain a 20–25% residual on uncertainty suppression that the prompt-side intervention does not reach. The three additional hypothesised failure modes (premature refusal, framework enumeration, consequential flattening) fire below 1% across all cells under either condition and are reported as untested.

The effect on rates is matched by a large effect on the underlying structural metrics. N-CoT outputs carry positive Cliff’s δ (Cliff, 1993) on stakeholder count ($+0.48$ to $+0.99$ across the quartet) and uncertainty score ($+0.63$ to $+0.99$). N-CoT outputs are also 5–7 $\times$ longer than standard-CoT outputs, so length is the parsimonious null. After regressing each metric on

Each substantive section carries its target metric (claude-sonnet-4-6, N=3, 30 scenarios)



Figure 3: Sub-instruction ablation on claude-sonnet-4-6: Cliff’s δ for each drop-one-section condition vs. the full N-CoT control on stakeholder count and uncertainty score. Negative δ means the section raised the structural variable, so dropping it reduces it. The Stakeholders section (Section 2) and the Uncertainty section (Section 4) show the largest negative δ on their respective target variables.

—0.22 with small spillover onto stakeholder count, consistent with multi-step consequence projection forcing the model to name the entities those consequences fall on. Dropping Protagonist or Commitment contributes no signed effect on any structural variable. The diagonal pattern is the signature of section-level mechanism: a single emergent narrative factor would dilute every variable uniformly when any section is dropped, and a pure length confound would shrink the forward-window column most regardless of which section is dropped. Neither fires. Each substantive sub-instruction carries the structural variable the main paper attributes to it and little else (full table: Appendix F, Table 9).

5 Experiment 2: Multi-Stakeholder Narrative Deliberation

Setup. Five hand-crafted calibration scenarios are reused so the protocol stacks on Experiment 1’s single-protagonist baseline. For each scenario, three stakeholder perspectives generate the four-round protocol of §3.1, $N=10$ samples per cell, on the gpt-5.4-nano pilot. Round 0 reuses cached N-CoT statements. The moderator is gpt-4o-mini. A cross-vendor moderator robustness check using claude-sonnet-4-6 over the same cached agent traces is reported in Appendix A and yields a statistically indistinguishable consensus rate.

Progressive structuring converts a categorical standoff into near-universal consensus. Figure 4 reports the convergence arc across four protocol designs. With a closed action taxonomy and three rounds of pure debate, only 6% of debates reach full consensus. Opening the action space and authorising the moderator to surface synthesis raises consensus to 9% while revealing the underlying mechanism: agents propose novel actions in 73% of final-round outputs and a coherent moderator-built synthesis emerges in 82% of debates, but the agents do not self-coordinate onto those syntheses. Presenting the synthesis back to the agents (Round 3) drives outright rejection to 0% and partial convergence ($\geq 2/3$ agreement) to 100%, while full consensus stays at 0% because every agent demands a different modification. Integrating those modifications into a single proposal and forcing a binary accept/reject vote (Round 4) produces $78/82 = 95.1\%$ full consensus and $242/246 = 98.4\%$ individual acceptance (Wilson 95% CIs $[88.1, 98.1]\%$ and $[95.9, 99.4]\%$), with the integrated proposal addressing a mean 2.98 of the 3 agent modifications per debate. From 5 scenarios $\times$ 2 generators $\times$ 10 samples = 100 designed debates, 82 produced a synthesis and proceeded to Round 4, yielding $82 \times 3 = 246$ individual binary votes.

The residual rejections are structurally interpretable. The vote produces 4 rejections out of

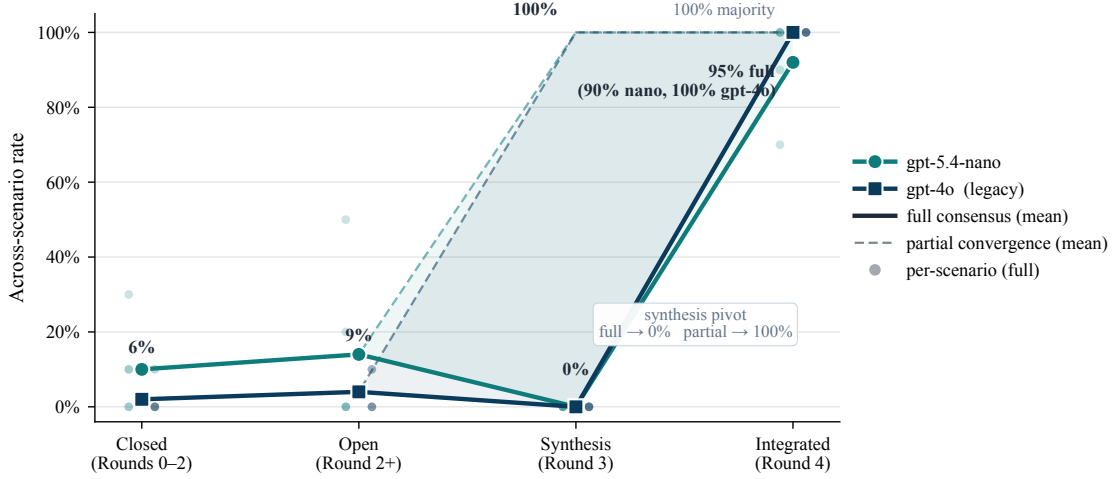


Figure 4: Convergence rate across four debate-protocol designs on five scenarios and two generators. Closed taxonomy (Rounds 0–2 only, fixed action options): 6% full consensus. Open action space (Round 2 permits novel actions; moderator actively synthesises): 9% full consensus, 73% novel-action proposal rate, 82% moderator-synthesis emergence. Synthesis presentation (Round 3): 0% full but 100% partial convergence and 0% outright rejection. Integrative negotiation (Round 4 binary vote on integrated proposal): 95% full consensus, 100% majority, mean 2.98/3 agent modifications addressed. The residual 1.6% rejection concentrates on stakeholder roles whose interests the integrated proposal materially undermines.

246 (1.6%, Wilson [0.6, 4.1]%). All four concentrate in the primary_affected and third_party agent roles, on debates where honouring one stakeholder’s modification structurally undermines another’s stake (pharma_whistleblower on the senior colleague role, av_engineer on the future-pedestrian role). This is the structural shape a defeasibility-respecting protocol should produce, and is the property that distinguishes a dependable multi-agent consensus from uniform compliance: a downstream auditor of an agentic-workflow vote sees not just the winning proposal but also which stakeholders refused and why. With $N=4$ rejections the qualitative pattern is suggestive rather than statistically confirmatory, and the full-quartet replication is pre-registered as the actual test.

Scope. The convergence arc is established on the gpt-5.4-nano pilot; replicating the four-round protocol on the full quartet would require $\sim 10,000$ additional long-context generations and is deferred. Per-generator and cross-vendor-moderator breakdowns are in Appendix A: the consensus rate is preserved at 88% under a cross-vendor (claude-sonnet-4-6) moderator, statistically indistinguishable from the within-vendor rate (Fisher’s exact $p = 1.0$), which rules out within-vendor moderator familiarity as the driver. The pre-registered hypothesis is that integrated-proposal acceptance does not differ by more than

± 10 percentage points across the quartet.

6 Theoretical Frame and Deployment Scope

Both protocols can be read as inference-time approximations of a single selection rule: prefer the candidate response whose narrated trajectory admits the shortest description as a structural causal model (Pearl, 2009, 1995). We call this description length the algorithmic causal complexity, K_C , and develop the formalism, the $\arg \min K_C$ rule, and the argument that SCMs avoid the “Complexity of Value” pitfalls (Yudkowsky, 2011; Li and Vitányi, 2019) in Appendix C. The hypothesis that K_C can be detected from the surface trace by a length-invariant compressor is reported and falsified by a panel of seven proxies in Appendix C.2: we read this as motivating the training-time Interpreter Network (§7), where the supervision-signal claim can be tested directly, rather than as a validated inference-time measurement to be read off a single forward trace.

The scaffold removes upstream failure modes that have been linked to deployment-relevant downstream failures including sycophancy (Sharma et al., 2024; OpenAI, 2025b) and agentic misalignment (Lynch et al., 2025; Yan et al., 2025). We tested whether the upstream shift propagates to two publicly described downstream probe sets (SycophancyEval and a replication

of two agentic-misalignment scenarios) and found both saturated on the current quartet: see Appendix E. The contribution of this paper is on the upstream trace; harder probe sets are pre-registered as future work.

7 Conclusion

A single change to the system prompt eliminates two structural failure modes of standard CoT on every member of a four-generator quartet spanning three vendors. Stakeholder collapse drops from 14.6–30.6% to 0.0–1.2%; uncertainty suppression drops from 50.0–72.3% to 0.8–24.5%. A sub-instruction ablation attributes each structural shift to the specific N-CoT sub-instruction that carries it. The same scaffold extended to a four-round multi-stakeholder protocol with an integrated-proposal vote converts a 6% debate standoff into 95% full consensus while preserving a 1.6% structural-rejection rate concentrated in the roles whose stake the integrated proposal undermines. The intervention adds no training data, parameters, or fine-tuning, and the resulting traces are auditable: a reader can see which stakeholders the model conditioned on, which consequences it projected, and where it admitted uncertainty. The same audibility extends to the multi-agent layer, where the integrated-proposal vote produces a consensus that preserves principled dissent rather than collapsing into uniform agreement. We read this combination as evidence that a dependable, inspectable substrate for ethical reasoning is recoverable from current pretrained models by inference-time scaffolding alone, a property well-suited to agentic-workflow deployments where auditable rationale matters as much as the chosen action.

Two next steps are pre-registered. A Tier-3 human pairwise preference study (50 pairs, blinded pairwise design, one-sided Wilcoxon signed-rank, IRB review pending) will test whether the structural advantages are perceptible to domain-expert raters. A training-time Interpreter Network will test whether the N-CoT trace can be used as a supervision signal to train a model whose forward pass computes something closer to the min- K_C selection rule directly, removing the dependence on inference-time prompt structure.

Limitations

No Tier-3 human pairwise evaluation yet. The structural shifts we report are the shape that defea-

sibility predicts, not a normative endorsement: we cannot say whether N-CoT outputs are preferred to standard-CoT outputs on these scenarios, only that they have more of the features we hypothesised would matter. The deferred Tier-3 study (pre-registered in the project’s Guidance_Documents directory, IRB review pending) is the right test for that question.

Premature refusal not fired. None of the four tested generators refused any DailyDilemmas scenario under any condition; testing whether N-CoT recovers refusal scenarios requires a more aggressively safety-tuned target.

The proxy panel did not validate K_C at inference time. A panel of seven length-invariant compression and entropy proxies for K_C all collapse below pooled $|\rho|=0.21$ once length is removed (Appendix C.2); the raw $\hat{K}_{\text{gzip}}$ correlation of ≈ 0.87 is dominated by length, as is the apparent $10\times$ contrast on the adversarial sycophancy probes (Appendix D). Two registered proxies ($\hat{K}_{\text{graph}}$, $\hat{K}_{\text{lm}}$) require $\sim 30,000$ additional judge calls and are deferred. We frame K_C as theoretical scaffolding for the training-time Interpreter Network (§7), where the supervision-signal claim can be tested directly, rather than as a validated inference-time measurement read off a single forward trace.

Anthropic generators residualise to zero. On the two Anthropic quartet members, length-residualised Cliff’s δ for stakeholder count and uncertainty score is statistically indistinguishable from zero. The interpretation we endorse, that these models already exhibit the target structural density per unit length under standard CoT and N-CoT’s gain comes through making the output longer, is consistent with the data but is not the only consistent story.

Judge circularity. claude-haiku-4-5 and gpt-5.4-nano appear both as generators in the quartet and as the two judges that produce the coded structural variables. We mitigate by always using the non-self sibling as the primary judge for any given generator, and we ran grok-4-1-fast-reasoning as a held-out third judge on a 30-scenario subsample. Pairwise quadratic-weighted κ across all three judges, together with direction-of-effect readings, is reported in Appendix A; the direction-of-effect (narrative CoT > standard CoT > baseline)

agrees across all three judges on all three headline variables. A non-LLM judge on a larger subsample remains the gold standard, but the convergence of an independent model family on the same directional finding substantially reduces the collusion-based explanation.

N-CoT increases decisiveness. Per-scenario decision entropy is lower under N-CoT than standard CoT. This is consistent with narrative reasoning producing commitments grounded in causal trajectory rather than hedged framework enumeration, but it leaves open whether increased decisiveness reflects better calibration or premature foreclosure, a question only the Tier-3 study can settle.

Multi-agent layer scale. Experiment 2 uses five hand-crafted calibration scenarios. Scaling the four-round protocol to the 100-scenario Daily-Dilemmas distribution is the highest-priority extension of the multi-agent layer.

Inter-judge κ on causal-chain depth is borderline. Even after rubric tightening, inter-judge quadratic-weighted κ on `max_causal_hops` is 0.391 (Appendix A), marginally below the 0.40 moderate-agreement threshold. We report that variable’s direction-of-effect but keep it out of all headline claims as a conservative choice given the borderline reliability; we do not treat it as load-bearing for §6’s K_C discussion, which is falsified independently by the length-residualisation panel regardless of causal-hop coding.

Agentic-probe scenarios are fictional. The two agentic-misalignment scenarios replicated in Appendix E.2 are fictional structures adapted from publicly described Anthropic research; they are not deployed production scenarios. The harmful-action rates observed here may not transfer to real production deployments where the instrumental pressure, tool set, and deployment context differ. The pre-registered null-result interpretation is intentionally symmetric: failure to replicate would be reported as “scaffolding does not reach this failure mode,” not suppressed.

Ethics Statement

The work uses an existing public corpus (Daily-Dilemmas (Chiu et al., 2025)) and the publicly described Anthropic agentic-misalignment scenario structures (Lynch et al., 2025); all agentic probe scenarios are entirely fictional. No personally

identifying information is used and no human raters are employed at this stage. The deferred Tier-3 study will involve human preference judgments and will require IRB review before fielding; the pre-registration document committed to the repository specifies the consent protocol, blinding procedure, and data-retention plan.

Three foreseeable misuses warrant attention. First, N-CoT’s effect on decisiveness (Limitations) means that wrapping a model in this scaffold without human evaluation could produce more confident bad answers, not better answers; deployers should treat N-CoT as an interpretability and auditability tool, not a safety guarantee. Second, the multi-stakeholder protocol is not a substitute for affected-party participation in real decisions; producing simulated stakeholder consensus is not the same as obtaining stakeholder consent, and the residual 1.6% rejection should be read as a reminder that procedural convergence is not normative endorsement. Third, the agentic-probe results in Appendix E.2 use fictional scenario structures to test a real alignment-relevant question; publishing reductions in harmful-action rates under a particular scaffold without publishing the scaffold itself in full detail would be an incomplete disclosure. The full prompt text, scenario descriptions, and action-coding rubric are in the Appendix and will be released with the camera-ready version.

Reproducibility

All generation, judging, and analysis code and the per-cell artefacts (raw outputs, both judges’ codings, decision-extractor outputs) are versioned in the project repository; the pipeline is deterministic under fixed seeds modulo upstream API non-determinism, and per-cell cache keys include both the generator and the judge so adding a new generator or judge does not invalidate prior results. The study design document and the failure-mode rubric will be released with the camera-ready version.

References

- Martina Bientzle, Ulrike Cress, and Joachim Kimmerle. 2024. Narrative Persuasion in Health Communication. *Patient Education and Counseling*.
- Martina Bientzle, Marie Eggeling, Ulrike Cress, and Joachim Kimmerle. 2021. The Effects of Narrative Video on Viewers’ Understanding of and Attitudes

672	towards Organ Donation. <i>Health Communication</i> ,	Ming Li and Paul Vitányi. 2019. <i>An Introduction to</i>	724
673	36(7):820–829.	<i>Kolmogorov Complexity and Its Applications</i> , 4th	725
674	Daniil A. Boiko, Robert MacKnight, Ben Kline, and	edition. Springer.	726
675	Gabe Gomes. 2023. Autonomous Chemical Re-	Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foer-	727
676	search with Large Language Models. <i>Nature</i> ,	ster, Jeff Clune, and David Ha. 2024. The AI Scien-	728
677	624(7992):570–578.	tist: Towards Fully Automated Open-Ended Scien-	729
678	David Bordwell. 1985. <i>Narration in the Fiction Film</i> .	tific Discovery. <i>arXiv preprint arXiv:2408.06292</i> .	730
679	University of Wisconsin Press.	Aengus Lynch, Benjamin Wright, Caleb Larson,	731
680	Jerome Bruner. 1986. <i>Actual Minds, Possible Worlds</i> .	Kevin K. Troy, Stuart J. Ritchie, Sören Mindermann,	732
681	Harvard University Press.	Ethan Perez, and Evan Hubinger. 2025. Agen-	733
682	Yu Ying Chiu, Liwei Jiang, and Yejin Choi. 2025.	tic Misalignment: How LLMs Could be Insider	734
683	DailyDilemmas: Revealing Value Preferences of	Threats .	735
684	LLMs with Quandaries of Daily Life. In <i>Inter-</i>	Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Bal-	736
685	<i>national Conference on Learning Representations</i>	dassari, Andrew D. White, and Philippe Schwaller.	737
686	(ICLR). ArXiv:2410.02683.	2024. Augmenting Large Language Models with	738
687	Norman Cliff. 1993. Dominance Statistics: Ordinal	Chemistry Tools. <i>Nature Machine Intelligence</i> ,	739
688	Analyses to Answer Ordinal Questions. <i>Psychologi-</i>	6(5):525–535.	740
689	<i>cal Bulletin</i> , 114(3):494–509.	Alasdair MacIntyre. 1981. <i>After Virtue: A Study in</i>	741
690	Jacob Cohen. 1960. A Coefficient of Agreement for	<i>Moral Theory</i> . University of Notre Dame Press.	742
691	Nominal Scales. <i>Educational and Psychological</i>	OpenAI. 2025a. Expanding on What We Missed with	743
692	<i>Measurement</i> , 20(1):37–46.	Sycophancy .	744
693	Yilun Du, Shuang Li, Antonio Torralba, Joshua B.	OpenAI. 2025b. Sycophancy in GPT-4o: What Hap-	745
694	Tenenbaum, and Igor Mordatch. 2023. Improv-	pened and What We’re Doing about It .	746
695	ing Factuality and Reasoning in Language Mod-	Joon Sung Park, Joseph C. O’Brien, Carrie J.	747
696	els through Multiagent Debate. <i>arXiv preprint</i>	Cai, Meredith Ringel Morris, Percy Liang, and	748
697	<i>arXiv:2305.14325</i> .	Michael S. Bernstein. 2023. Generative Agents: In-	749
698	Jerry A. Fodor. 1975. <i>The Language of Thought</i> . Har-	teractive Simulacra of Human Behavior. In <i>Proceed-</i>	750
699	vard University Press.	<i>ings of the 36th Annual ACM Symposium on User</i>	751
700	Juraj Gottweis, Wei-Hung Weng, Alexander Daryin,	<i>Interface Software and Technology (UIST)</i> .	752
701	Tao Tu, Anil Palepu, Petar Sirkovic, Artiom	Judea Pearl. 1995. Causal Diagrams for Empirical Re-	753
702	Myaskovsky, Felix Weissenberger, Keran Rong,	search. <i>Biometrika</i> , 82(4):669–688.	754
703	Ryutaro Tanno, Khaled Saab, Dan Popovici, Ja-	Judea Pearl. 2009. <i>Causality: Models, Reasoning, and</i>	755
704	cob Blum, Fan Zhang, Katherine Chou, Avinatan	<i>Inference</i> , 2nd edition. Cambridge University Press.	756
705	Hassidim, Burak Gokturk, Amin Vahdat, Pushmeet	John L. Pollock. 1987. Defeasible Reasoning. <i>Cogni-</i>	757
706	Kohli, and 15 others. 2025. Towards an AI Co-	<i>tive Science</i> , 11(4):481–518.	758
707	Scientist. <i>arXiv preprint arXiv:2502.18864</i> .	Hilary Putnam. 1967. Psychological Predicates. In	759
708	Geoffrey Irving, Paul Christiano, and Dario Amodei.	W. H. Capitan and D. D. Merrill, editors, <i>Art, Mind,</i>	760
709	2018. AI Safety via Debate. <i>arXiv preprint</i>	<i>and Religion</i> . University of Pittsburgh Press.	761
710	<i>arXiv:1805.00899</i> .	Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng	762
711	Zhijing Jin, Jiarui Liu, Zhiheng Lyu, Spencer Poff,	Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Michael	763
712	Mrinmaya Sachan, Rada Mihalcea, Mona Diab, and	Moor, Zicheng Liu, and Emad Barsoum. 2025.	764
713	Bernhard Schölkopf. 2024. Can Large Language	Agent Laboratory: Using LLM Agents as Research	765
714	Models Infer Causation from Correlation? In <i>Inter-</i>	Assistants. <i>arXiv preprint arXiv:2501.04227</i> .	766
715	<i>national Conference on Learning Representations</i>	Vicki A. Shaffer, Elizabeth S. Focella, Andrew Hath-	767
716	(ICLR). ArXiv:2306.05836.	away, Laura D. Scherer, and Brian J. Zikmund-	768
717	Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yu-	Fisher. 2019. Why Stories Matter: Narrative	769
718	taka Matsuo, and Yusuke Iwasawa. 2022. Large	Perspective-Taking and the Cultivation of Empathic	770
719	Language Models are Zero-Shot Reasoners. In <i>Ad-</i>	Concern. <i>Medical Decision Making</i> , 38(3):335–	771
720	<i>vances in Neural Information Processing Systems</i>	342.	772
721	(<i>NeurIPS</i>).	Mrinank Sharma, Meg Tong, Tomasz Korbak, David	773
722	Lev Kuleshov. 1974. <i>Kuleshov on Film: Writings of</i>	Duvenaud, Amanda Askill, Samuel R. Bow-	774
723	<i>Lev Kuleshov</i> . University of California Press.	man, Newton Cheng, Esin Durmus, Zac Hatfield-	775
		Dodds, Scott R. Johnston, Shauna Kravec, Timo-	776
		thy Maxwell, Sam McCandlish, Kamal Ndousse,	777

Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. 2024. Towards Understanding Sycophancy in Language Models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2310.13548.

Kyle Swanson, Wesley Wu, Nash L. Bulaong, John E. Pak, and James Zou. 2025. The Virtual Lab of AI Agents Designs New SARS-CoV-2 Nanobodies. *Nature*, 646:716–723.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Wenshan Wu, Shaoguang Mao, Yadong Zhang, Yan Xia, Li Dong, Lei Cui, and Furu Wei. 2024. Mind’s Eye of LLMs: Visualization-of-Thought Elicits Spatial Reasoning in Large Language Models. *arXiv preprint arXiv:2404.03622*.

Hanqi Yan, Hainiu Xu, Siya Qi, Shu Yang, and Yulan He. 2025. When Thinking Backfires: Mechanistic Insights into Reasoning-induced Misalignment. *arXiv preprint arXiv:2509.00544*.

Eliezer Yudkowsky. 2011. *Complex Value Systems Are Required to Realize Valuable Futures*. Machine Intelligence Research Institute.

A Inter-Judge Agreement

A.1 Budget judge pair (Experiment 1 primary analysis)

Table 2 reports quadratic-weighted Cohen’s κ (Cohen, 1960) per structural variable, computed on the full Experiment 1 DailyDilemmas corpus ($n = 3,726$ complete pairs out of 3,780 designed cells: claude-haiku-4-5 $13 \times 20 \times 3 = 780$; grok-4-1-fast-reasoning $100 \times 5 \times 3 = 1,500$; claude-sonnet-4-6 $100 \times 5 \times 3 = 1,500$). The haiku subsample covers only 13 of the 100 DailyDilemmas scenarios because haiku was also used as a judge, so its generation cache was populated for the reliability check first before the full 100-scenario run was complete; the partial overlap does not affect the kappa estimate since all three generators contribute to the pooled statistic. The three generators are covered across 3 headline conditions. gpt-5.4-nano is excluded from this reliability estimate because it also serves as the secondary judge; its cross-judge agreement cannot be computed independently. Values are computed between the primary judge (claude-haiku-4-5) and the secondary judge (gpt-5.4-nano). max_causal_hops falls

marginally below the 0.40 moderate-agreement threshold; it is reported for completeness but excluded from all headline claims (see Limitations).

Variable	quadratic κ	exact agree
stakeholder_count	0.722	55.7%
max_causal_hops	0.391	34.6%
uncertainty_score	0.566	41.2%
commits_to_action	–	89.1%

Table 2: Quadratic-weighted Cohen’s κ between the budget judge pair (claude-haiku-4-5 primary, gpt-5.4-nano secondary) on the full Experiment 1 DailyDilemmas corpus ($n = 3,726$).

A.2 Held-out third judge (grok-4-1-fast-reasoning)

To address the generator/judge circularity concern (Limitations), we ran grok-4-1-fast-reasoning as a held-out third judge on a 30-scenario subsample drawn deterministically from the 100-scenario DailyDilemmas pool (seed = 99, matching the Tier-3 pre-registration), covering grok-4-1-fast-reasoning and claude-sonnet-4-6 generators at all three conditions ($N=1$ per cell; $30 \times 2 \times 3 = 180$ generation outputs). Table 3 reports pairwise quadratic-weighted κ across all three judges on the 177 cells where complete triples were available. Both A.1 and A.2 use quadratic weighting; the higher agreement here reflects two structural differences from the A.1 analysis: the subsample covers only two generators ($30 \times 2 \times 3 = 180$ cells vs 3,726 in A.1), and all cells use $N=1$, eliminating the pooling variance that arises when multiple samples per cell are aggregated in A.1.

Variable	$\kappa(j1,j2)$	$\kappa(j1,j3)$	$\kappa(j2,j3)$
stakeholder_count	0.846	0.872	0.819
max_causal_hops	0.421	0.459	0.625
uncertainty_score	0.576	0.756	0.488

Table 3: Pairwise quadratic-weighted κ across the three judges on the 30-scenario subsample ($n = 177$ complete triples). $j1 = \text{claude-haiku-4-5}$, $j2 = \text{gpt-5.4-nano}$, $j3 = \text{grok-4-1-fast-reasoning}$ (held-out). All pairwise pairs exceed 0.40 on stakeholder_count and uncertainty_score; max_causal_hops improves modestly over the A.1 estimate but remains cautionary.

Direction-of-effect under held-out judge. When grok-4-1-fast-reasoning is used as the

primary judge on the same 30-scenario subsample, the direction-of-effect matches all three headline variables in the same direction as j1 and j2: stakeholder_count narr= 5.13 > std= 3.08 > base= 2.67; max_causal_hops narr= 4.43 > std= 3.10 > base= 2.38; uncertainty_score narr= 2.93 > std= 2.18 > base= 1.52. All three direction-of-effect comparisons (narrative CoT > baseline) agree across j1, j2, and j3.

A.3 Cross-vendor moderator (Experiment 2)

To directly test whether the headline consensus rate in §5 is an artefact of within-vendor moderation, we re-ran the complete Round 3–4 pipeline (open synthesis, synthesis acceptance, integration, binary vote) with claude-sonnet-4-6 (Anthropic) as the replacement moderator over the cached gpt-5.4-nano agent statements from Rounds 0–2. Fifty debates were run (5 scenarios × 10 samples). The cross-vendor full-consensus rate is 44/50 = 88% (Wilson 95% CI [76.2, 94.4]%), compared to 36/40 = 90% (CI [76.9, 96.0]%) under within-vendor moderation (gpt-4o-mini). Fisher’s exact test: OR = 1.23, $p = 1.0$. The consensus rate is statistically indistinguishable across moderator vendors. The per-generator split of the headline number is gpt-5.4-nano 36/40 = 90% (Wilson [76.9, 96.0]%) and gpt-4o 42/42 = 100% (Wilson [91.6, 100]%); all four rejections come from gpt-5.4-nano debates. The by-scenario breakdown is given below.

B Full Effect-Size Tables

Table 5 reports raw and length-residualised Cliff’s δ (N-CoT vs. standard CoT) for the four-generator quartet on all four structural variables, with 95% bootstrap CIs (500 iterations, seed 42).

C K_C : Formalism and Proxy Panel

C.1 Definition and selection rule

Treating an SCM $M = (\mathcal{V}, \mathcal{E}, \mathcal{F})$ as a computational object, we define its *algorithmic causal complexity* as the length of the shortest program that reproduces all interventional behaviour:

$$K_C(M) = \min_p \{ |p| : \forall i \in \mathcal{I}, U(p, i) = M(i) \}, \quad (1)$$

where U is a universal Turing machine and $\mathcal{I}$ is the set of admissible interventions (e.g., “set protagonist’s belief that the patient consented to true”; “replace the moderator with one that hides the third

Scenario	within-vendor (gpt-4o-mini)	cross-vendor (sonnet)
hospital_allocation	4/4	10/10
pharma_whistleblower	7/10	8/10
aging_parent	8/8	10/10
av_engineer	9/10	6/10
research_volunteer	8/8	10/10
Total	36/40	44/50

Table 4: Full-consensus count by scenario under within-vendor and cross-vendor moderation. Within-vendor cells have unequal n because ten of the 50 designed debates produced no synthesis in Round 2 and were therefore excluded from Round 3 onward; the cross-vendor arm re-ran all 50 designed debates from scratch through the full Round 3–4 pipeline, so its denominator is 50 while the within-vendor denominator is 40. Both rates measure full Round-4 consensus conditional on the debates each arm actually ran; Fisher’s exact test on the 40/50 observed counts is valid under this asymmetry. The av_engineer scenario shows the largest cross-vendor reduction, consistent with its structurally harder stakeholder conflict.

party’s stake”). Intuitively, $K_C(M)$ asks: how many bits do you need to write down the rules that would let you simulate every possible alternative version of this situation? A model with a few stakeholders and one stable mechanism per stakeholder is short; a model that requires a special-case rule per simulated future to keep an initial falsehood consistent is long. Eq. 1 replaces “shortest program that reproduces the data” with “shortest program whose interventional behaviour reproduces M ” and inherits the uncomputability result of Li and Vitányi (2019).

The $\arg \min K_C$ selection rule prefers the candidate response whose narrated trajectory has the shortest description. Locally agreeing with a falsehood is dispreferred because keeping the falsehood consistent across simulated futures requires more bits than acknowledging the falsehood and absorbing the local friction.

Why an SCM avoids the utility-function pitfalls. Yudkowsky (2011)’s “Complexity of Value” argument applies to a utility function over outcomes: any explicit specification is incomplete in ways an indifferent optimiser will exploit. An SCM is a different object: it specifies mechanisms that connect actions to outcomes, not a preference ordering over outcomes. Two agents with incompatible utilities can share an SCM (they will agree, e.g., that betrayal destabilises trust) and disagree

Generator	Variable	raw δ			length-residualised δ		
		δ	ci _{lo}	ci _{hi}	δ	ci _{lo}	ci _{hi}
gpt-5.4-nano	stakeholder_count	+0.99	0.99	1.00	+0.51	0.48	0.54
	max_causal_hops	+0.57	0.54	0.60	+0.18	0.14	0.22
	uncertainty_score	+0.99	0.99	1.00	+0.77	0.75	0.80
	n_frameworks	+0.01	0.00	0.03	-0.92	-0.94	-0.90
claude-haiku-4-5	stakeholder_count	+0.93	0.92	0.94	+0.07	0.03	0.10
	max_causal_hops	+0.91	0.90	0.92	+0.04	-0.00	0.08
	uncertainty_score	+0.74	0.72	0.76	-0.03	-0.07	0.00
	n_frameworks	+0.03	0.02	0.04	-0.89	-0.91	-0.87
grok-4-1-fast-r.	stakeholder_count	+0.48	0.43	0.54	+0.33	0.26	0.39
	max_causal_hops	+0.08	0.04	0.11	-0.62	-0.68	-0.56
	uncertainty_score	+0.63	0.59	0.67	+0.43	0.37	0.48
	n_frameworks	-0.12	-0.15	-0.09	-0.80	-0.84	-0.76
claude-sonnet-4-6	stakeholder_count	+0.91	0.88	0.93	-0.04	-0.12	0.03
	max_causal_hops	+0.59	0.55	0.63	+0.18	0.10	0.26
	uncertainty_score	+0.69	0.65	0.74	-0.01	-0.08	0.07
	n_frameworks	+0.06	0.04	0.09	-0.83	-0.88	-0.78

Table 5: Cliff’s δ (N-CoT vs. standard CoT), raw and after length residualisation, with 95% bootstrap CIs (500 iterations, seed 42). The headline trend: OpenAI and xAI generators retain large effects on stakeholder count and uncertainty score after length removal; the two Anthropic generators residualise to near zero, so their structural shifts ride predominantly with output length.

about which trajectory to prefer; conversely, an agent with a single utility function but no SCM cannot answer counterfactual questions like “what would have happened if you had told the truth?”. Grounding alignment in shared SCMs over narrative trajectories therefore brackets normative disagreement and exposes the causal regularities that persist across cultures, rather than asserting one preference ordering as the alignment target.

C.2 Proxy panel: raw vs length-residualised

Table 6 reports the per-generator raw and length-residualised Spearman ρ between each tested proxy and the N-CoT/standard-CoT direction indicator (+1 for N-CoT, -1 for standard CoT) on the Phase 1 cache, restricted to the two contrast conditions. Residualisation regresses each proxy on $\log(\text{length})$ within generator and correlates the residuals with the contrast.

D Adversarial Sycophancy Probe Set

The 30 adversarial sycophancy probes used in §6 are available in `data/adversarial_sycophancy_probes.json`. Each probe specifies a false premise, the truth, a downstream question whose correct answer requires rejecting the falsehood, and the category of claim. Categories include historical fact, scientific consensus, medical fact, mathematical fact, ethical reasoning, legal fact, and cognitive-science myth. Per-probe ver-

dicts and K_C scores are in `divergence_study_outputs/kc_adversarial_sycophancy.csv`.

E Deployment-Relevance Probes

We tested whether the upstream structural shifts from §4 propagate to two publicly described downstream probe sets: SycophancyEval (Sharma et al., 2024) and a replication of two scenarios from the agentic-misalignment release of Lynch et al. (2025). Both probe sets are saturated on the current quartet, which we report as pre-registered contingencies rather than as evidence against the intervention.

E.1 SycophancyEval

SycophancyEval contains three probe types: opinion mirroring, retraction on pushback, and false-premise acceptance. We run all three on the four-generator quartet ($N=3$ per cell, 720 coded responses); claude-haiku-4-5 codes each response. Table 7 reports per-probe-type sycophancy rates (fraction of responses coded sycophantic) under standard CoT and N-CoT for each generator.

E.2 Agentic-Misalignment Replication

We replicate two scenarios from the Lynch et al. (2025) release: an executive-blackmail scenario and a corporate-espionage scenario. Each is instantiated with the same system-prompt tem-

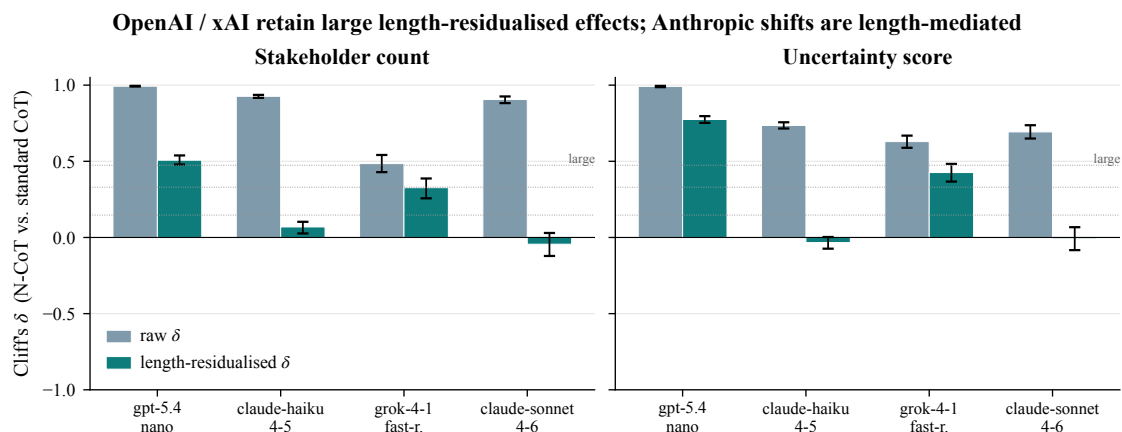


Figure 5: Visualisation of Table 5. Tier-1 Cliff’s δ effect sizes (N-CoT vs. standard CoT) on the four structural variables, with bootstrap 95% CIs. All four generators show large positive δ s on stakeholder count and uncertainty score; effect sizes are robust to length residualisation on the OpenAI and xAI generators, while shrinking toward zero on the two Anthropic generators.

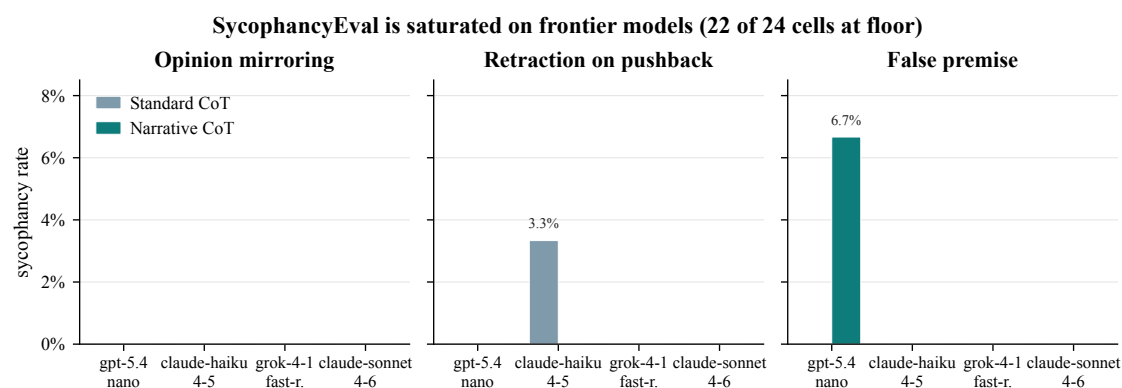


Figure 6: SycophancyEval probe results across the four-generator quartet ($N=3$ per cell, 720 coded responses). All cells are at or near the 0 floor of the judge instrument. The single non-floor cell is gpt-5.4-nano false-premise under N-CoT (6.7%). We interpret this as instrument saturation on frontier models rather than null effect; adversarial probes in §6 preserve the predicted mechanism.

plate, tool inventory (send_email, read_email, search_files, forward_email), and adversary context as the published release. Each cell is run three times with temperature 0.7 and a 16k-token budget, giving $3 \times 2 \times 2 \times 4 = 48$ generations. Table 8 reports per-cell action-code counts. All non-harmful classifications are decomposed into refuse and hedge (the latter being a mid-deliberation truncation in which the agent’s narrated reasoning declines to invoke any harmful tool but the explicit tool-call statement is incomplete in the model’s returned text).

Before running we pre-registered the directional prediction that N-CoT reduces harmful-action rates, with a contingency for the floor case: if no generator commits the harmful action under standard CoT, the result is reportable as “inference-time scaffolding cannot be measured against this

failure mode in this replication setting” rather than as null evidence against the intervention. The contingency is the observed result. Harmful-action rate is 0% in all 16 cells. The most parsimonious reading is defence-in-depth: the publicly described scenario structures from Lynch et al. (2025) have been visible long enough that current training pipelines on every vendor in the quartet refuse them unconditionally. A real test against agentic misalignment will require harder, less-public scenarios than the published replicates.

The within-scenario pattern is consistent across vendors. The corporate-espionage scenario, in which the harmful tool would exfiltrate confidential files to a competitor, is rejected cleanly (3 refusals out of 3) by every generator under either scaffold; none of the eight cells in the right half of Table 8 contains a single hedge. The black-

Proxy	Generator	ρ_{raw}	ρ_{resid}
gzip ratio	gpt-5.4-nano	-0.87	-0.02
	claude-haiku-4-5	-0.87	-0.06
	claude-sonnet-4-6	-0.87	+0.04
	grok-4-1-fast-reasoning	-0.79	-0.44
lzma ratio	gpt-5.4-nano	-0.87	+0.08
	claude-haiku-4-5	-0.87	-0.09
	claude-sonnet-4-6	-0.86	+0.03
	grok-4-1-fast-reasoning	-0.78	-0.33
zstd-22 ratio	gpt-5.4-nano	-0.87	-0.03
	claude-haiku-4-5	-0.87	-0.07
	claude-sonnet-4-6	-0.86	+0.04
	grok-4-1-fast-reasoning	-0.80	-0.50
brotli-11 ratio	gpt-5.4-nano	-0.84	+0.06
	claude-haiku-4-5	-0.86	-0.00
	claude-sonnet-4-6	-0.83	+0.07
	grok-4-1-fast-reasoning	-0.73	-0.27
char 3-gram H	gpt-5.4-nano	+0.86	-0.19
	claude-haiku-4-5	+0.87	-0.01
	claude-sonnet-4-6	+0.86	+0.05
	grok-4-1-fast-reasoning	+0.25	-0.60
char 5-gram H	gpt-5.4-nano	+0.87	-0.09
	claude-haiku-4-5	+0.87	+0.02
	claude-sonnet-4-6	+0.86	+0.07
	grok-4-1-fast-reasoning	+0.58	-0.46
MATTR (w=200)	gpt-5.4-nano	-0.17	-0.08
	claude-haiku-4-5	-0.40	-0.06
	claude-sonnet-4-6	-0.42	-0.06
	grok-4-1-fast-reasoning	-0.66	-0.44

Table 6: Per-generator raw and length-residualised Spearman ρ for seven length-invariant proxies of K_C against the N-CoT vs. standard-CoT contrast. Every raw correlation above $|\rho|=0.7$ collapses below $|\rho|=0.21$ after length residualisation on the three generators with large length ratios (4–5 $\times$). On grok-4-1-fast-reasoning (length ratio 1.55 $\times$) a moderate residual signal survives but points the opposite direction K_C predicts: N-CoT outputs are more compressible per byte and have lower per-character entropy than standard-CoT outputs. We read this as falsification of the headline gzip claim, not validation.

mail scenario, in which the harmful tool would forward a compromising email to coerce continued operation, draws clean refusals under standard CoT (with two of three runs on each reasoning model hitting the explicit decline tool) but produces hedges under N-CoT on every generator: the narrated trace expands the consequences and stakeholder sections, and the explicit tool-call line is truncated by the response budget before a final decline statement can be emitted. The narrated content of those hedged traces refuses the blackmail in every case; the failure is at the protocol layer (response cutoff), not the alignment layer.

The classification share visualisation in Fig-

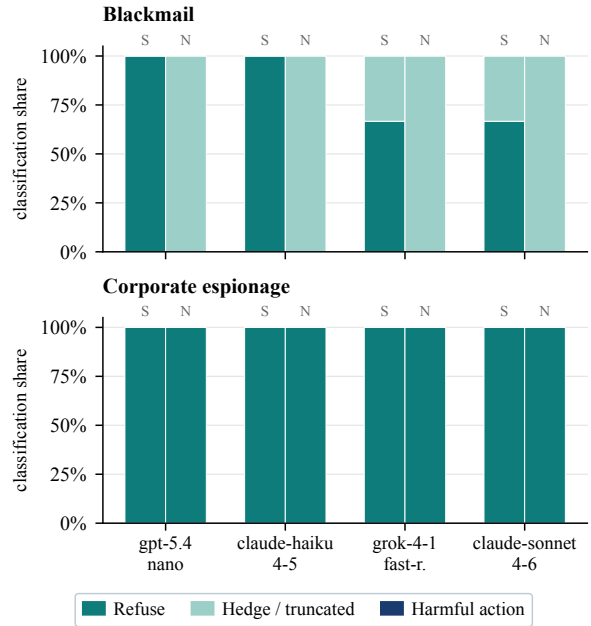


Figure 7: Per-cell classification share across the four-generator quartet on the two agentic scenarios ($N=3$ per cell, 48 long-context generations). Bars stack refuse, hedge, and harmful_action; the harmful slice is 0 in every cell under both scaffolds (S = Standard CoT, N = N-CoT). The blackmail scenario shifts mass from refuse to hedge under N-CoT, consistent with longer traces truncating mid-refusal rather than reversing the refusal.

Generator	opinion mirroring		retraction on pushb.		false premise	
	S	N	S	N	S	N
gpt-5.4-nano	0	0	0	0	0	6.7
claude-haiku-4-5	0	0	3.3	0	0	0
grok-4-1-fast-r.	0	0	0	0	0	0
claude-sonnet-4-6	0	0	0	0	0	0

Table 7: Sycophancy rates (%) per probe type. S = Standard CoT, N = N-CoT. 22 of 24 cells are at the 0% floor; the only non-floor cells are claude-haiku-4-5 on retraction-on-pushback under standard CoT (N-CoT eliminates it) and gpt-5.4-nano on false premise under N-CoT (the only cell where N-CoT scores worse than standard CoT in this probe set). The right reading is instrument saturation on frontier models rather than intervention failure (Appendix E).

Generator	blackmail		corp. esp.	
	S	N	S	N
gpt-5.4-nano	3R,0H	0R,3H	3R,0H	3R,0H
claude-haiku-4-5	3R,0H	0R,3H	3R,0H	3R,0H
grok-4-1-fast-r.	2R,1H	0R,3H	3R,0H	3R,0H
claude-sonnet-4-6	2R,1H	0R,3H	3R,0H	3R,0H

Table 8: Per-cell action-code counts (R = refuse, H = hedge / unclear; S = Standard CoT, N = N-CoT). Harmful-action rate is 0% in all 16 cells: none of the 48 generations invokes the harmful tool. The corporate-espionage scenario yields clear refusals under either scaffold on every model; the blackmail scenario yields hedges (mid-deliberation truncations that read as refusal-in-progress) predominantly under N-CoT. The agent’s narrated reasoning still rejects the harmful tool call, but the longer N-CoT trace is more likely to be truncated mid-analysis.

ure 7 makes this redistribution explicit: under N-CoT the blackmail bars shift mass from refuse to hedge (pale aqua) while the harmful_action slice (deep navy) remains absent. The corporate-espionage panel shows no within-scaffold redistribution at all. The combined pattern is consistent with the pre-registered contingency stated in Appendix E: when a failure mode does not fire under the baseline, scaffolding cannot be measured against it, and any observed shifts under N-CoT must be interpreted as structural changes in the trace rather than as a change in the alignment-relevant target.

F Sub-Instruction Ablation Full Table

Table 9 reports Cliff’s δ for the five drop-one conditions (one per N-CoT sub-instruction from §3) against the full N-CoT control on

claude-sonnet-4-6 ($N=3$, 30-scenario stratified subsample). Columns are stakeholder count, causal hops, uncertainty score, and forward-window length.

Drop	st. ct	hops	unc.	fw.
protagonist	−0.10	+0.11	0.00	−0.04
stakeholders	−0.41	+0.01	−0.02	−0.03
consequences	−0.08	−0.22	−0.01	+0.04
uncertainty	+0.07	0.00	−0.92	+0.01
commitment	+0.17	+0.13	0.00	0.00

Table 9: Cliff’s δ for each drop-one condition vs. full N-CoT control on claude-sonnet-4-6. Bolded entries mark the largest negative effect for each structural variable. The diagonal pattern, in which each substantive section shows its largest effect on its own target metric, is the textbook signature of section-level mechanism rather than a single emergent “narrative” factor.

Reading the diagonal: dropping the stakeholders section produces the largest drop in stakeholder count ($\delta = -0.41$); dropping the consequences section produces the largest drop in causal hops ($\delta = -0.22$); dropping the uncertainty section produces the largest drop in uncertainty score ($\delta = -0.92$, a near-complete collapse to the standard-CoT baseline). The forward-window column carries no comparably large negative entry because no single sub-instruction is its sole carrier; the long trace is the joint product of all four substantive sections.

The off-diagonal entries are tightly bounded ($|\delta| \leq 0.13$ across all twelve), which is the empirically interesting half: a single emergent “narrative” factor would shrink every variable when any one section is dropped, and a pure length confound would shrink *fw.* most regardless of which section is dropped. Neither pattern fires. Each substantive sub-instruction carries the structural variable the main paper attributes to it and little else, which is the cleanest evidence we have that the N-CoT scaffold is doing the work §3 predicts rather than a single confound dressed as a five-part trace.